SMART INDIA HACKATHON 2026

Problem Statement ID - 26231

Problem Statement Title -Digital Companion for Field Drug Testing

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - K26234

Team Name - HexTech



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PROPOSED SOLUTION

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Problem

India: 1,20,010 NDPS cases registered in 2023 (NCRB, Crime in India 2023)

- Officers mainly judge the kit's colour change by looking at it, so lighting and personal judgement can affect the result.
- There is no standard colour value to compare the test result with.
- The test does not always leave a clear digital record of who performed it, where, and when, making it harder to verify later.

Solution

- **Check the Test** - The app photographs the kit next to a ₹10 colour card and says POSITIVE, NEGATIVE or NOT SURE.
- **Record the Test** - Saves the result with the photo fingerprint, time, GPS and Inspector ID, signed for a QR check.
- **Manage the Records** - The admin dashboard manages Inspectors, test records, NOT SURE reviews and reports.

Test Kit → Take Photo → Check Colour → Get Result → Save Proof → Admin Review

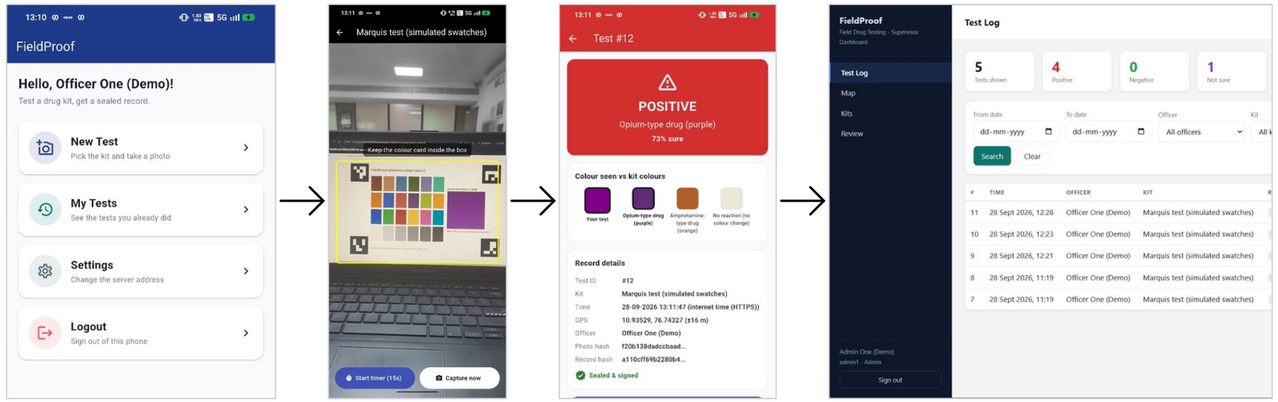
Prototype Demo Video: Prototype Video Link

Novelty

Not just a colour reader. A field result anyone can check later.

- **Handles Lighting Variations** - 24 known colours on a ₹10 card correct the light before the kit colour is read.
- **Handles Unclear Results** - Blurry, shiny or in-between photos get NOT SURE with the reason, never a guess.
- **Creates Verifiable Evidence** - Photo fingerprint, GPS, internet time and Inspector ID are signed and checked by QR.
- **Supports Multiple Kits** - New kit colour profiles are added from the Admin website without changing the app.
- **Local Language Support** - Officers can use the app in Tamil.

Prototype Flow - real screenshots of our working prototype



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TECHNICAL APPROACH

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System Architecture

Officer's phone (Flutter app)
- Login and pick kit
- Kit read timer
- Camera with card guide
- Blur check on phone
- GPS and mock flag
- Photo SHA-256

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Colour engine (server, OpenCV)
- Find card (4 corners)
- Blur and glare check
- Fix colours (24 known)
- CIELAB and ΔE 2000
- Pick closest kit colour

POSITIVE | NEGATIVE | NOT SURE + reason

↓

Sealed record (server)
- Internet time (NTP, HTTPS)
- 22 fields sealed
- SHA-256 record hash
- Ed25519 signature
- Chain to last record

↓

Storage
- PostgreSQL: users, tests, kit profiles, reviews
- Photo store, PDF with QR and BSA s.63 page

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Admin Dashboard
- Test log and CSV
- Map
- Kit editor
- NOT SURE review
- QR check page

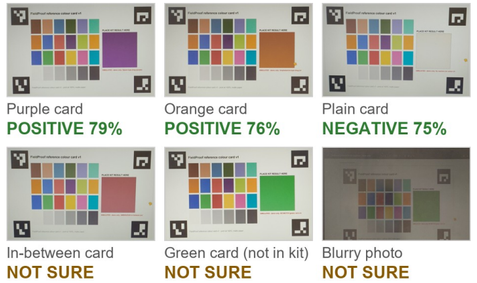
Everything shown is built. No AI model, on purpose: colour maths that can be explained with numbers.

Measured Results

- **Testing** - 37 computer-made photos in 5 lights and 25 real phone scans.
- **Accuracy** - 25/25 computer photos right. Real scans: 14 right, 11 NOT SURE, 0 wrong.
- **Colour Check** - ΔE ≤ 14 and a 4-point lead, else NOT SURE. The % shown is a match score, not a probability.
- **Speed** - Server check 2.5 s (5 s max). Whole test about 16 seconds.
- **Tamper Test** - All 4/4 changed records were caught.

Kit colours are printed swatches on a screen, 1 phone. Small test, not proof. Real kits need an FSL trial (TRL 4).

Prototype Validation Using Real Phone Scans (29 Sep)



Purple card — POSITIVE 79%
Orange card — POSITIVE 76%
Plain card — NEGATIVE 75%
In-between card — NOT SURE
Green card (not in kit) — NOT SURE
Blurry photo — NOT SURE

1 phone, 1 room, card on a laptop screen. We checked each card by eye.

All 19 real scans

Card	POS	NEG	NOT SURE
Positive (8)	5	0	3
Plain (4)	0	4	0
Trick (7)	0	0	7

0 wrong answers
All 9 clear scans correct
All 7 trick cards said NOT SURE
3 blurry photos, app asked to retake

Technology Stack

- Flutter (Dart)
- FastAPI (Python)
- OpenCV + NumPy
- CIELAB + ΔE2000
- PostgreSQL + SQLAlchemy
- React + Vite
- JWT + bcrypt
- SHA-256
- Ed25519
- NTP
- Leaflet + OpenStreetMap
- ReportLab

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FEASIBILITY AND VIABILITY

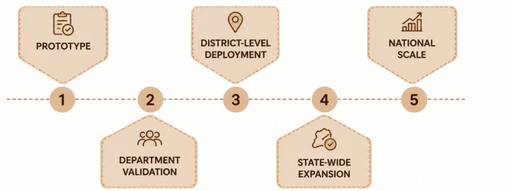
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Feasibility & Sustainability

Legal: first check only, the FSL lab confirms it.
Privacy: no face, fingerprint or suspect name, only a sample code.

- **Technical** - Full chain works today: app, server, database and dashboard (TRL 4).
- **User-Friendly** - One photo, result in about 16 s (tested by our team so far).
- **Low Cost** - Existing Android phone, a reusable ₹10 colour card and open-source tools.
- **Operational** - Simple photo-based workflow. Officer trial is part of the pilot.
- **Scalable** - New kit profiles from the Admin website, after FSL calibration.
- **Sustainable** - State police IT cell runs it, FSL keeps kit colours right (proposed).

Phased Implementation & Scalability



Pilot Plan (proposed)

- **Where** - 1 district in Tamil Nadu, 20 officers, 3 months, nearest FSL.
- **Check 2 things apart** - app vs an expert reading the same kit (target 95%+), and kit vs FSL lab.
- **Size** - 60+ tests per kit, so 0 wrong shows error under 5%.
- **Pass if** - NOT SURE under 10%, every record passes the QR check.
- **Cost** - about ₹9,200, cards and server only (estimate).

Potential Challenges & Strategies

Real-world problem	What our app does	Status
Lighting can change the colour seen by the phone.	24 known colours on the card correct the colour before analysis.	Built
Blurry or shiny photos can make the result unclear.	Detects it, says NOT SURE with the reason and asks for a retake.	Built
Some colours may fall between two results.	Shows NOT SURE instead of guessing. Supervisor can only order a retest, sample goes to FSL.	Built
Kit colour changes with time.	Each kit has a read time, the app timer takes the photo on time.	Built
Photos or records can be edited later.	Photo fingerprint, signature and chain catch any change (4/4 in our test).	Built
Someone photographs a screen or a staged colour.	In-app camera only today. Screen-photo check and phone integrity check next.	Planned
Different kits and phones show colours differently.	Kit colour settings and colour fixed in every photo. Real kits need FSL calibration.	Built

Cost & Deployment Plan

After the pilot, the cost can come from the state police or NCB IT budget. Server about ₹36,000 a year (estimate).

COST	HOSTING	MAINTENANCE	DEPLOYMENT	SCALE
Existing phone, ₹10 card, open-source software	NIC or state data centre in India (to confirm)	Security updates, FSL keeps kit colours right	Pilot with police and FSL, then scale across locations	New test kits added through the website, no app update

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IMPACT AND BENEFITS

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Inspector Benefits

- Same reading in any light
- Less manual colour checking
- NOT SURE when unclear
- Easy digital test records
- Quick access to past test records

Organisation Benefits

- Proof of every test: time, place, officer & photo
- Any change to a record is caught
- PDF with QR and BSA s.63 page
- Easy search, map and CSV export
- All test data in one system

QR-Based Record Verification

Judges can scan it on their own phone. For the demo the QR opens a public HTTPS check page (verify only, nothing else is open).

FieldProof - Field Test Record

Presumptive field result — not a substitute for laboratory confirmatory testing.

POSITIVE
Opium-type drug (purple) | 76% sure



Scan to check this record

Test number	11
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Saved at (UTC)	2026-09-28T06:58:23.601Z (time from: internet time (HTTPS))
Phone time	2026-09-28T12:28:21+05:30 (difference: -2.6 s)
GPS location	10.935299, 76.743282 (±14.791999816894531 m)
Officer	Officer One (Demo) (user id 1)
Device ID	android-1e42a9c170701656ac8a22b0ca44a6e0
Kit	Marquis test (simulated swatches) [DEMO-MARQUIS-SIM], profile version 1
Lab sample ID	-
Review status	NOT_NEEDED

FieldProof · Record check

✓ Record is genuine
The photo and details were not changed after they were saved.

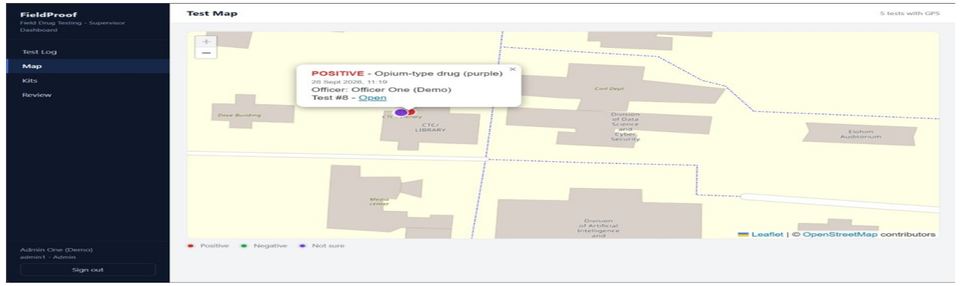
What we checked

- ✓ The photo is the same as when it was taken
- ✓ The saved details match their fingerprint (hash)
- ✓ The FieldProof server's digital signature is valid
- ✓ The record is correctly linked to the record before it
- ✓ The result, officer and kit shown are the ones that were signed

Record details

Test number	11
Result	POSITIVE
What the kit showed	Opium-type drug (purple)
How sure	76%
Kit	Marquis test (simulated swatches) (version 1)
Officer	Officer One (Demo)
Saved at (UTC)	2026-09-28T06:58:23.601Z (internet time (HTTPS))
GPS	10.93530, 76.74328
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Photo fingerprint (SHA-256)	294b35454c7e963c7e0f939b4f0e9cd10a279e69fca62634d61ea030d98a9e57

Map - Admin can monitor area-wise test results on the map.



Evidence Report

Every PDF report has a filled BSA section 63 page (photo hash, phone, time, GPS) for the officer and expert to sign. Open a full sample report (PDF, 2 pages)

Future Scope

- **Phone Security** - sign each test with the phone's security chip.
- **Lab Comparison** - FSL trial with real kits, compare with lab results.
- **Screen-photo Check** - catch photos of a screen, phone integrity check.
- **Police Systems** - apply to NIC or MHA for the e-Sakshya and CCTNS link.
- **Regional Languages** - Tamil is done, Hindi and voice help next.

Offline Mode (Planned, month 1-2) - Test and photo hash saved and signed on the phone, synced when the network returns. Duplicates saved once.

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RESEARCH AND REFERENCES

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What they found	How we use it	Research
IEEE EMBC 2016 Phone camera colours change with light and phone model, and reference colours can help handle this.	Our ₹10 colour card fixes colours in every test	[1] H. Karisen and T. Dong, "Illumination and device independence for colorimetric detection of urinary biomarkers with smartphone," in Proc. 38th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), Orlando, FL, USA, 2016, pp. 5184-5187, doi: 10.1109/EMBC.2016.7591895.
IEEE TPS-ISA 2024 Digital evidence must maintain integrity and show how it was handled.	Photo fingerprint + signature + chained log	[2] S. Nath, K. Summers, J. Baek, and G.-J. Ahn, "Digital evidence chain of custody: Navigating new realities of digital forensics," in Proc. IEEE 6th Int. Conf. Trust, Privacy Secur. Intell. Syst. Appl. (TPS-ISA), Washington, DC, USA, 2024, pp. 11-20, doi: 10.1109/TPS-ISA62245.2024.00012.
Forensic Science International 2014 A mobile phone camera can measure drug-test colours using RGB values.	Our app reads the test colour as a number	[3] A. Choodum, K. Parabun, N. Klawach, N. Nic Daeid, P. Kanatharana, and W. Wongniramaikul, "Real time quantitative colourimetric test for methamphetamine detection using digital and mobile phone technology," Forensic Sci. Int., vol. 235, pp. 8-13, Feb. 2014, doi: 10.1016/j.forsciint.2013.11.018.
Analytical Chemistry 2016 Reading the colour by eye is subjective, and phone image analysis can give an objective reading.	App gives POSITIVE / NEGATIVE / NOT SURE	[4] S. T. Krauss et al., "Objective method for presumptive field-testing of illicit drug possession using centrifugal microdevices and smartphone analysis," Anal. Chem., vol. 88, no. 17, pp. 8689-8697, Sep. 2016, doi: 10.1021/acs.analchem.6b01982.
Digital Discovery 2023 Computer vision can quantify and compare colour changes that may look similar by eye.	App shows photo colour next to kit colours	[5] N. Bugeja et al., "Teaching old presumptive tests new digital tricks with computer vision for forensic applications," Digital Discovery, vol. 2, no. 4, pp. 1143-1151, 2023, doi: 10.1039/D3DD00066D.

What we add: earlier work shows a phone can read test colours [1][3][4][5]. We add the NOT SURE rule, a signed record anyone can check with a QR, and a BSA section 63 page.

Scan Smart. Verify Clearly. Protect Evidence.

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TEST RESULTS IN DETAIL

Backup slide for judges' questions

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All Tests in One Table

Test	n	Setup	Result
Computer-made photos	37	5 lights: day, bulb, tube light, dim, street lamp. Rules were tuned on these	25/25 good right, 12/12 bad NOT SURE
Real scans, 28 Sep	6	Card shown on a phone or laptop screen	5 right, 1 NOT SURE, 0 wrong
Real scans, 29 Sep	19	8 positive, 4 plain, 7 trick cards. 1 room, window light	9 right, 10 NOT SURE, 0 wrong
All real scans	**25**	1 phone (Nothing A015), our team only	**14 right, 11 NOT SURE, 0 wrong**
Stress test	171	19 real photos x 9 computer-made changes (dark, yellow bulb, tilt, shake, grain)	70 right, 101 NOT SURE, 0 wrong. Sharp photos 69/72 right
Tamper test	5	Copies of saved records, 4 changes and 1 untouched	4/4 changes caught, original passed
Speed	18	Server, 6 real photos x 3 runs	2.5 s typical, 5 s max

What 0 wrong means

0 wrong in 14 real answers is a small test. By the rule of three the true error rate could still be up to about 21%.

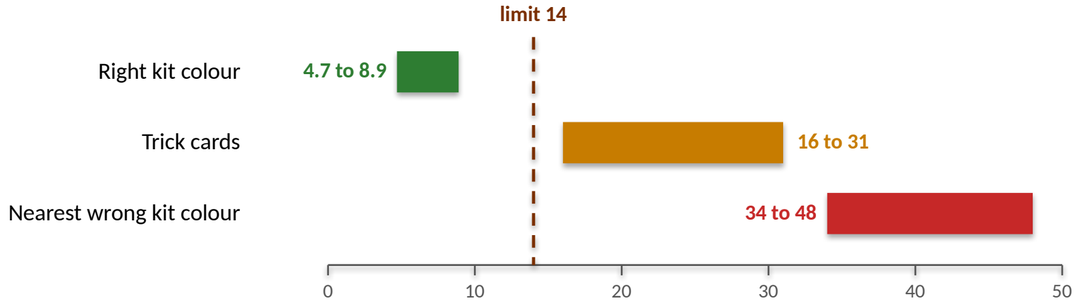
To show under 5% we need 60+ tests with 0 wrong, so the pilot asks for 60+ tests per kit.

Photo 7 on 28 Sep was rejected when the card fit limit was lower. We quote that live result, not a re-run.

What we do not claim

- Accuracy on real reagents (not tested yet)
- Which drug it is (the FSL lab confirms)
- Court approval. It supports the BSA s.63 certificate, the officer and expert still sign
- Results on other phones or with officers (next step)

Colour Gap on Real Scans (ΔE 2000)



Right colours sat well under 14 and wrong kit colours far above it. The closest trick card was only about 2 points over the limit, so each real kit gets its own limit from FSL data.

Card fit (average error left on the 24 known colours after correction): 5.9 to 8.7 on real photos of a screen, limit 9. Under 1 on computer-made photos. Paper card test is next.

Why the 29 Sep Scans Said NOT SURE

- **3 positive cards** - photo was blurry (records 13, 14, 20), the app asked for a retake.
- **7 trick cards** - 3 matched no kit colour, 4 were stopped earlier by photo checks (1 blurry, 2 card not found, 1 card fit).

Tamper Test on Our Saved Records (29 Sep)

Test #10
`prev 2ece5c80`
`hash 6b1b4e7f`

→

Test #11
`prev 6b1b4e7f`
`hash 03dd9f0c`

→

Test #12
`prev 03dd9f0c`
`hash a110cff6`

Ed25519 signature of test 11
`JRIdu6ERgadC+2n6chKfUStp...`
Photo SHA-256 (signed in record)
`294b35454c7e963c7e0f939b...`

What we changed (test #11)	Check page	Failed check
Nothing (original)	GENUINE	none, all 5 passed
1 byte of the photo	TAMPERED	photo
Result inside sealed copy	TAMPERED	hash, signature, fields
Result in database row	TAMPERED	fields
The signature	TAMPERED	signature

Tried on copies of our saved records (29 Sep 2026). Hash, signature and previous hash are stored in the database.

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SECURITY AND FSL TRIAL

Backup slide for judges' questions

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Threat Model (what is built, what is open)

Threat	What stops it today	Next step	Status
Photo or record changed later	Photo SHA-256 on phone and server, signed record, chain, QR check	Supervisor counter-seal, officer DSC	Built, 4/4 caught
Photo of a screen or staged colour	In-app camera only, no gallery upload	Screen-photo (moire) check, Play Integrity	Open
Fake GPS app	Android mock-location flag sealed in the record	Play Integrity device check	Partly
Wrong phone clock	Server takes internet time, phone time and the gap are also saved	Outside time-stamp (RFC 3161)	Built
Insider with the server key	Key only on the server, public key is published	Key in a hardware box (HSM), chain head copied outside daily	Open
Server key file lost	Server makes a new key, old records then fail the check	Stop and alert instead, keep a key backup	Fix planned
Public QR page shows officer name and exact GPS	Page is verify only, nothing else is open	Hide the name, round off GPS	Fix planned
Test done with no network	Not saved today	Offline queue built together with phone chip signing	Planned
One chain for a whole state	One chain today (prototype)	Chain per device or a daily batch root	Planned

Legal hooks: BSA 2023 s.63 (certificate for electronic records, with hash), BNSS 2023 s.105 (recording of search and seizure), NDPS SSSD Rules 2022, DPDP Act 2023 s.8(5) (keep data safe).

FSL Trial to Set Each Kit's Limit

1. **Real reagent**
   FSL runs each kit on known samples, 3+ kit lots, photo at the kit's read time and again later to see colour drift.

2. **Phones and light**
   Paper card, 3+ phones, daylight, yellow bulb, tube light and street lamp.

3. **Measure**
   Card fit on paper, spread of the right colour, distance to the nearest wrong colour.

4. **Set limits**
   ΔE limit and lead for each kit from that gap, with a NOT SURE band in between. Freeze them before testing.

5. **Test fresh samples**
   Samples not used for tuning, compared with the FSL lab. Report agreement with a range, not one number.

New kit, step by step

FSL calibrates the kit, admin adds the kit profile (it gets a version number), a small field check, then officers can use it. Kit lot and expiry will be saved in each test record (planned).